

- 3** An average-sized human chromosome contains a DNA molecule with about 300 million nucleotides, arranged in pairs.

(a) Describe the structure of a nucleotide pair.

(3)

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- (b) Before a cell can divide, the DNA molecule has to replicate. This takes place in the S phase of the cell cycle.

Human DNA replicates at a rate of 50 nucleotides per second.

Calculate how long it would take, to the nearest hour, for 150 million nucleotides to replicate.

Assume that replication starts at one end of the molecule and continues to the other end.

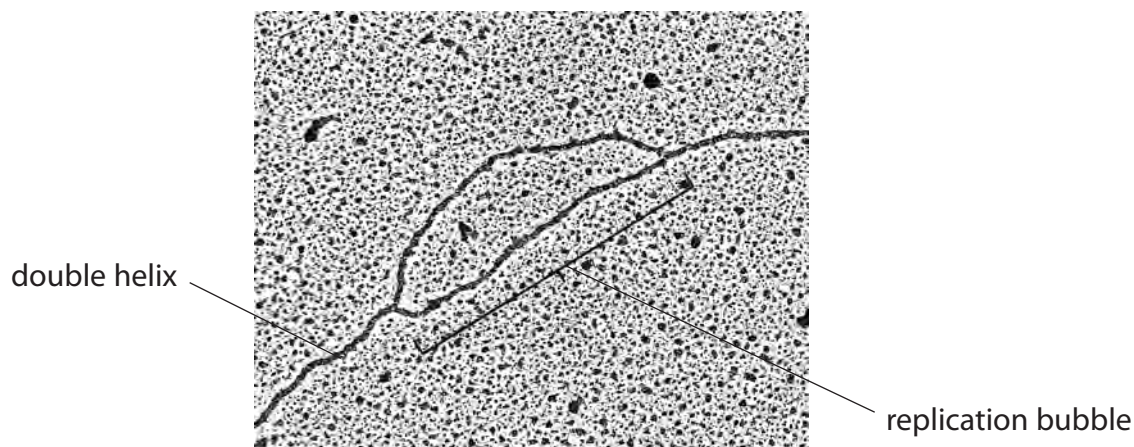
(2)

Answer hours



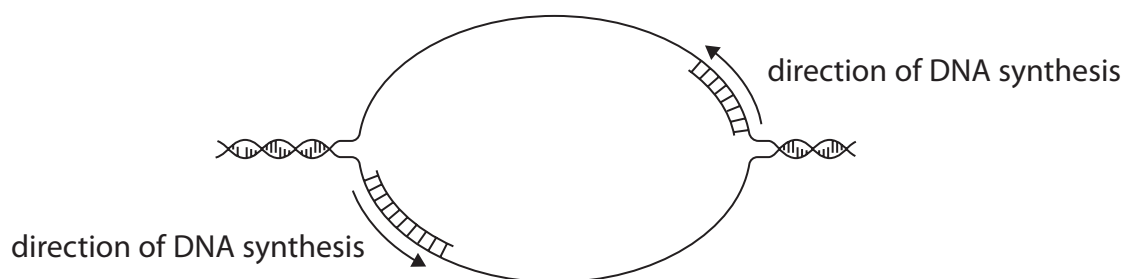
- (c) In human cells, replication of DNA occurs at several sites along the molecule. These sites are called replication bubbles.

The photograph shows a replication bubble.



(Source: © Dr Gopal Murti/Science Photo Library)

The diagram shows how the DNA is replicated in one replication bubble.



- (i) Explain the role of DNA polymerase in a replication bubble.

Use the information in the diagram and your own knowledge to support your answer.

(2)

- (ii) In human cells, S phase lasts about 10 hours.

Suggest why each DNA molecule is replicated using many replication bubbles.

(3)

(Total for Question 3 = 10 marks)

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